



Juan Vallejo

Blockchain / Web3 Developer

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PROFESSIONAL SUMMARY

Full-stack Web3 developer with hands-on experience across DeFi, DEX, and CEX-style systems: built an atomic swap-and-stake DeFi flow on Ethereum (1inch Aqua SwapVM + Lido) for an ETHGlobal hackathon, and an autonomous crypto trading bot integrated with a centralized exchange (Binance Spot via CCXT) with strict risk controls. Comfortable across the stack — Solidity-adjacent tooling (ethers.js, viem, Hardhat), smart-contract-calling backends, and production-grade Node.js/NestJS APIs — plus 1.5+ years shipping backend and full-stack systems professionally. Active member of ETH Cali, the local Web3/blockchain hackathon community. Strong fundamentals in async processing, security-first design (JWT/RBAC, zero-withdrawal key scoping), and working fast in small, high-context teams.

TECHNICAL SKILLS

BLOCKCHAIN & WEB3

ethers.js viem Hardhat 1inch Aqua SDK / SwapVM Ethereum Mainnet Fork Testing Lido Staking Integration
Wallet / TWA Integrations ABI Encoding

EXCHANGE & TRADING SYSTEMS (CEX)

CCXT Binance Spot API Order Execution Risk Management (SL/TP) Backtesting Market Regime Detection

BACKEND & APIS

Node.js NestJS Express Next.js RESTful APIs WebSockets Webhooks

PROGRAMMING LANGUAGES

TypeScript JavaScript Python C# SQL

DATABASES & ORMS

PostgreSQL MongoDB SQLite Prisma Drizzle ORM

ASYNCH, BOTS & MESSAGING

Redis BullMQ Socket.io python-telegram-bot Telegram Mini Apps

SECURITY & AUTH

JWT RBAC Zero-Withdrawal Key Scoping Zod Validation bcrypt

DEVOPS & TOOLS

Docker Docker Compose Git GitHub Linux (Ubuntu) Postman

AI-ASSISTED DEVELOPMENT

Claude Code

FEATURED WEB3 PROJECTS

SwipeFi — Atomic DeFi Zap & Yield (ETHGlobal Hackathon)

1inch Aqua SDK SwapVM viem ethers.js Hardhat Mainnet Fork Lido Next.js Telegram Mini App

- ▶ Built a Telegram Mini App targeting the 1inch Aqua/SwapVM track: collapses swap-approve-stake into a single atomic on-chain transaction (USDC → ETH via Aqua liquidity → stETH via Lido's `submit()`), reverting fully if either leg fails.
- ▶ Implemented the SwapVM compiler (`compileAquaSwap`) wiring `AquaProtocolContract`, strategy-hash calculation, and `executeAtomic()` batch construction directly against the official `@1inch/aqua-sdk`, with all calldata ABI-encoded via viem.
- ▶ Validated the full swap-and-stake flow against a local Ethereum Mainnet Fork (Hardhat) before any testnet/mainnet exposure.

github.com/juanvallejo01 — ETHGlobal hackathon submission

Crypto Agent — Autonomous CEX Trading Bot (Binance Spot)

Python · CCXT · Flask · SQLite · Docker Compose · Telegram Bot API

- ▶ Built an autonomous quant trading bot for Binance Spot (SMA crossover + RSI, with an ML confirmation filter and market-regime detection) sized for small live accounts.
- ▶ Implemented strict risk controls: mandatory SL/TP on every position, a circuit breaker after consecutive losses, and API keys scoped with zero-withdrawal permissions end-to-end — no code path can move funds off-exchange.
- ▶ Shipped a Flask monitoring dashboard and a Telegram bot (/status /balance /pause /resume) for live control and alerts, plus a backtesting/parameter-optimization pipeline (backtest.py, optimize_config.py), deployed via Docker Compose.

github.com/juanvallejo01

Urbanix — Multi-Role SaaS Backend

NestJS 11 · Prisma · PostgreSQL · Redis · WebSockets · AWS S3

- ▶ Architected a 25+ module NestJS backend (finance, reservations, incidents, notifications) with JWT-secured REST APIs and real-time WebSocket channels for a production multi-role SaaS platform.

Client project — Codeliance

PROFESSIONAL EXPERIENCE

Backend Developer — Codeliance

Feb 2025 – Present

- ▶ Design and own backend architecture for production systems spanning residential management, sports club administration, and social automation — 25+ module NestJS platforms with PostgreSQL, Redis, and real-time channels.
- ▶ Automate administrative and documentary workflows across multiple client platforms, working directly with clients from requirements to deployment.

Backend Developer — TechNova Solutions

Apr 2024 – Nov 2024

- ▶ Designed RESTful APIs with Node.js, Express, and Prisma over PostgreSQL for a residential-management SaaS application.
- ▶ Automated internal processes with Python and AutoHotKey and integrated Google Workspace workflows.

EDUCATION

Computer Engineering (In Progress)

2023 – Present

Universidad Santiago de Cali, Colombia — Member, Comba Research & Development Incubator

CERTIFICATIONS

- ▶ **Databases:** Bases de Datos: Generalidades y Sistemas de Gestión — SENA (40h, 2025)
- ▶ **Programming:** C# — Ademass Training & Development · Python — Ademass Training & Development
- ▶ **Cybersecurity & Networking:** Ethical Hacker — Cisco Networking Academy (Verified) · Introduction to Cybersecurity — Cisco Networking Academy (Verified)

ACHIEVEMENTS & LEADERSHIP

- ▶ Active Member, ETH Cali — Web3 & blockchain hackathon community (2026 – Present); competed in the 1inch Aqua/SwapVM track at ETHGlobal with SwipeFi.
- ▶ Vice President, IEEE Computer Society — Universidad Santiago de Cali (2025 – 2026)

LANGUAGES

Spanish — Native

English — B2 (Upper Intermediate)